

- University of Jos

Department of Electrical/Electronics Engineering

EEE 303-2020: ELECTROMAGNETIC FIELDS AND WAVES 4

Faraday's Law of Induction

10.1 Faraday's Law of Induction

The electric fields and magnetic fields considered up to now have been produced by stationary charges and moving charges (currents), respectively. Imposing an electric field on a conductor gives rise to a current which in turn generates a magnetic field. One could then inquire whether or not an electric field could be produced by a magnetic field. In 1831, Michael Faraday discovered that, by varying magnetic field with time, an electric field could be generated. The phenomenon is known as electromagnetic induction. Figure 10.1.1 illustrates one of Faraday's experiments.

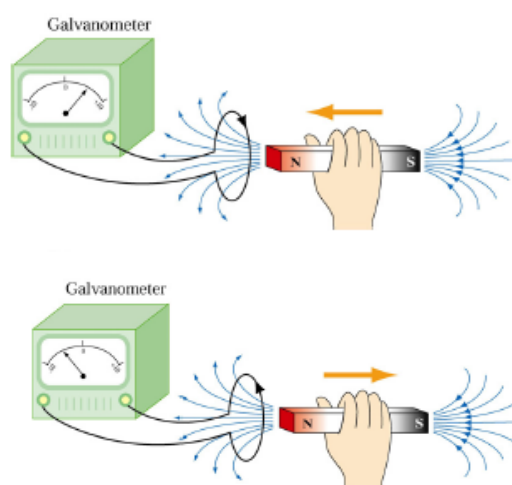


Figure 10.1.1 Electromagnetic induction

Faraday showed that no current is registered in the galvanometer when bar magnet is stationary with respect to the loop. However, a current is induced in the loop when a relative motion exists between the bar magnet and the loop. In particular, the galvanometer deflects in one direction as the magnet approaches the loop, and the opposite direction as it moves away.

Faraday's experiment demonstrates that an electric current is induced in the loop by changing the magnetic field. The coil behaves as if it were connected to an emf source. Experimentally it is found that the induced emf depends on the rate of change of magnetic flux through the coil.

10.1.1 Magnetic Flux

Consider a uniform magnetic field passing through a surface S , as shown in Figure 10.1.2 below:

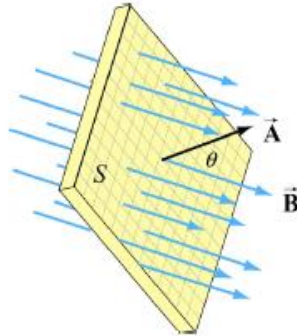


Figure 10.1.2 Magnetic flux through a surface

Let the area vector be $\vec{A} = A\hat{n}$, where A is the area of the surface and $\hat{n}$ its unit normal. The magnetic flux through the surface is given by

$$\Phi_B = \vec{B} \cdot \vec{A} = BA \cos \theta \quad (10.1.1)$$

$$\Phi_B = \iint_S \vec{B} \cdot d\vec{A} \quad (10.1.2)$$

The SI unit of magnetic flux is the weber (Wb):

$$1 \text{ Wb} = 1 \text{ T} \cdot \text{m}^2$$

Faraday's law of induction may be stated as follows:

The induced emf ε in a coil is proportional to the negative of the rate of change of magnetic flux:

$$\varepsilon = -\frac{d\Phi_B}{dt} \quad (10.1.3)$$

For a coil that consists of N loops, the total induced emf would be N times as large:

$$\varepsilon = -N \frac{d\Phi_B}{dt} \quad (10.1.4)$$

Combining Eqs. (10.1.3) and (10.1.1), we obtain, for a spatially uniform field $\vec{B}$,

$$\varepsilon = -\frac{d}{dt}(BA \cos \theta) = -\left(\frac{dB}{dt}\right)A \cos \theta - B\left(\frac{dA}{dt}\right) \cos \theta + BA \sin \theta \left(\frac{d\theta}{dt}\right) \quad (10.1.5)$$

Thus, we see that an emf may be induced in the following ways:

(i) by varying the magnitude of $\vec{B}$ with time (illustrated in Figure 10.1.3.)



Figure 10.1.3 Inducing emf by varying the magnetic field strength

(ii) by varying the magnitude of $\vec{A}$, i.e., the area enclosed by the loop with time (illustrated in Figure 10.1.4.)

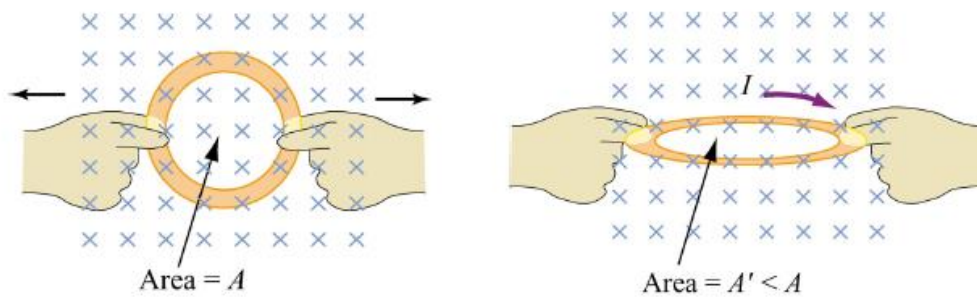


Figure 10.1.4 Inducing emf by changing the area of the loop

(iii) varying the angle between $\vec{B}$ and the area vector $\vec{A}$ with time (illustrated in Figure 10.1.5.)



Figure 10.1.5 Inducing emf by varying the angle between $\vec{B}$ and $\vec{A}$.

10.1.2 Lenz's Law

The direction of the induced current is determined by Lenz's law:

The induced current produces magnetic fields which tend to oppose the change in magnetic flux that induces such currents.

To illustrate how Lenz's law works, let's consider a conducting loop placed in a magnetic field. We follow the procedure below:

1. Define a positive direction for the area vector $\vec{A}$.
2. Assuming that $\vec{B}$ is uniform, take the dot product of $\vec{B}$ and $\vec{A}$. This allows for the determination of the sign of the magnetic flux Φ_B .
3. Obtain the rate of flux change $d\Phi_B / dt$ by differentiation. There are three possibilities:
3. Obtain the rate of flux change $d\Phi_B / dt$ by differentiation. There are three possibilities:

$$\frac{d\Phi_B}{dt}: \begin{cases} > 0 \Rightarrow \text{induced emf } \varepsilon < 0 \\ < 0 \Rightarrow \text{induced emf } \varepsilon > 0 \\ = 0 \Rightarrow \text{induced emf } \varepsilon = 0 \end{cases}$$

4. Determine the direction of the induced current using the right-hand rule. With your thumb pointing in the direction of $\vec{A}$, curl the fingers around the closed loop. The induced current flows in the same direction as the way your fingers curl if $\varepsilon > 0$, and the opposite direction if $\varepsilon < 0$, as shown in Figure 10.1.6.



Figure 10.1.6 Determination of the direction of induced current by the right-hand rule

In Figure 10.1.7 we illustrate the four possible scenarios of time-varying magnetic flux and show how Lenz's law is used to determine the direction of the induced current I .

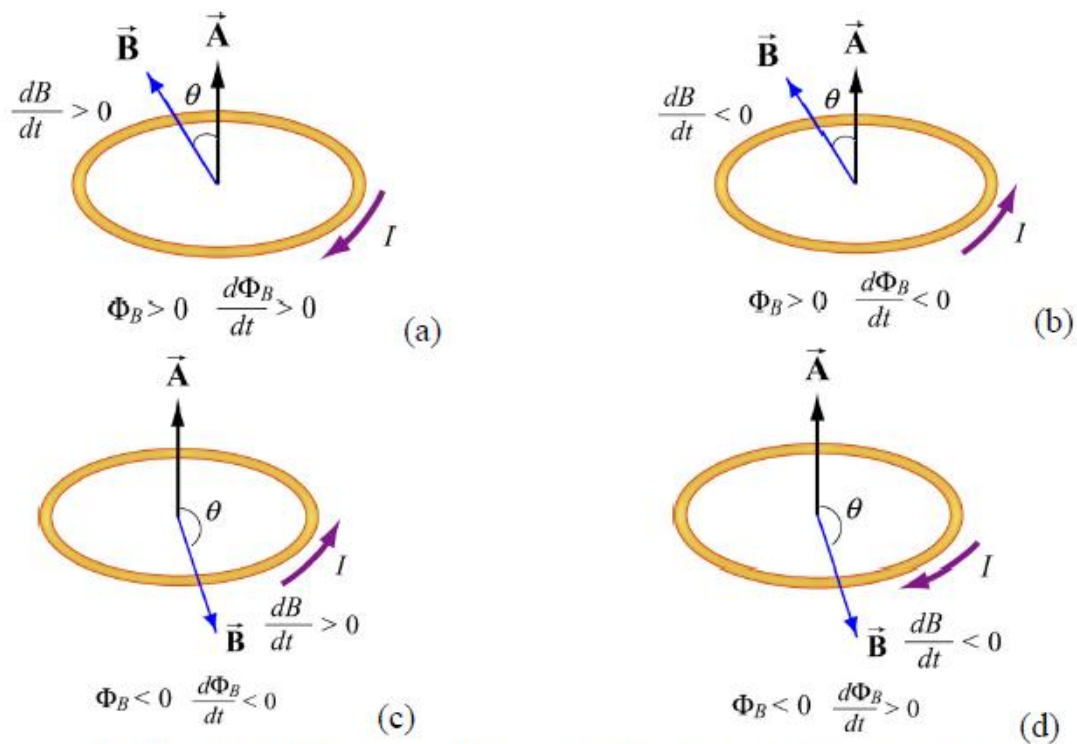


Figure 10.1.7 Direction of the induced current using Lenz's law

The above situations can be summarized with the following sign convention:

Φ_B	$d\Phi_B / dt$	$\mathcal{E}$	I
+	+	−	−
	−	+	+
−	+	−	−
	−	+	+

The positive and negative signs of I correspond to a counterclockwise and clockwise currents, respectively.

10.2 Motional EMF

Consider a conducting bar of length l moving through a uniform magnetic field which points into the page, as shown in Figure 10.2.1. Particles with charge $q > 0$ inside experience a magnetic force $\vec{F}_B = q\vec{v} \times \vec{B}$ which tends to push them upward, leaving negative charges on the lower end.

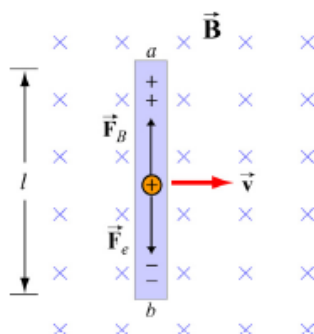


Figure 10.2.1 A conducting bar moving through a uniform magnetic field

The separation of charge gives rise to an electric field $\vec{E}$ inside the bar, which in turn produces a downward electric force $\vec{F}_e = q\vec{E}$. At equilibrium where the two forces cancel,

we have $qvB = qE$, or $E = vB$. Between the two ends of the conductor, there exists a potential difference given by

$$V_{ab} = V_a - V_b = \varepsilon = El = Blv \quad (10.2.1)$$

Since ε arises from the motion of the conductor, this potential difference is called the motional emf. In general, motional emf around a closed conducting loop can be written as

$$\varepsilon = \oint (\vec{v} \times \vec{B}) \cdot d\vec{s} \quad (10.2.2)$$

where $d\vec{s}$ is a differential length element.

where $\tau = mR / B^2 l^2$. Upon integration, we obtain

$$v(t) = v_0 e^{-t/\tau} \quad (10.2.11)$$

Thus, we see that the speed decreases exponentially in the absence of an external agent doing work. In principle, the bar never stops moving. However, one may verify that the total distance traveled is finite.

10.3 Induced Electric Field

In Chapter 3, we have seen that the electric potential difference between two points A and B in an electric field $\vec{E}$ can be written as

$$\Delta V = V_B - V_A = -\int_A^B \vec{E} \cdot d\vec{s} \quad (10.3.1)$$

When the electric field is conservative, as is the case of electrostatics, the line integral of $\vec{E} \cdot d\vec{s}$ is path-independent, which implies $\oint \vec{E} \cdot d\vec{s} = 0$.

Faraday's law shows that as magnetic flux changes with time, an induced current begins to flow. What causes the charges to move? It is the induced emf which is the work done per unit charge. However, since magnetic field can do not work, as we have shown in Chapter 8, the work done on the mobile charges must be electric, and the electric field in this situation cannot be conservative because the line integral of a conservative field must vanish. Therefore, we conclude that there is a non-conservative electric field $\vec{E}_{nc}$ associated with an induced emf:

$$\varepsilon = \oint \vec{E}_{nc} \cdot d\vec{s} \quad (10.3.2)$$

Combining with Faraday's law then yields

$$\oint \vec{E}_{nc} \cdot d\vec{s} = -\frac{d\Phi_B}{dt} \quad (10.3.3)$$

The above expression implies that a changing magnetic flux will induce a non-conservative electric field which can vary with time. It is important to distinguish between the induced, non-conservative electric field and the conservative electric field which arises from electric charges.

10.4 Generators

One of the most important applications of Faraday's law of induction is to generators and motors. A generator converts mechanical energy into electric energy, while a motor converts electrical energy into mechanical energy.

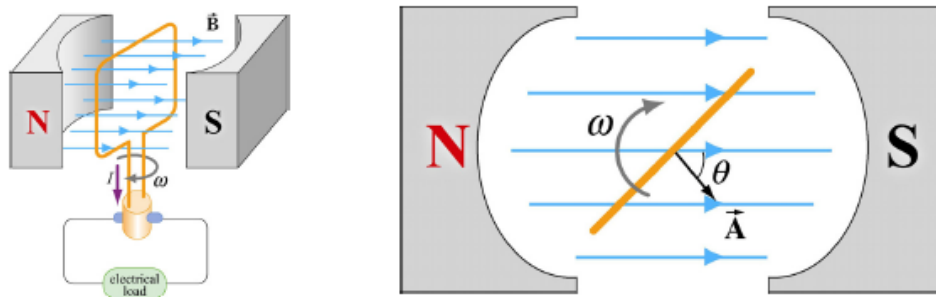


Figure 10.4.1 (a) A simple generator. (b) The rotating loop as seen from above.

Figure 10.4.1(a) is a simple illustration of a generator. It consists of an N -turn loop rotating in a magnetic field which is assumed to be uniform. The magnetic flux varies with time, thereby inducing an emf. From Figure 10.4.1(b), we see that the magnetic flux through the loop may be written as

$$\Phi_B = \vec{\mathbf{B}} \cdot \vec{\mathbf{A}} = BA \cos \theta = BA \cos \omega t \quad (10.4.1)$$

The rate of change of magnetic flux is

$$\frac{d\Phi_B}{dt} = -BA\omega \sin \omega t \quad (10.4.2)$$

Since there are N turns in the loop, the total induced emf across the two ends of the loop is

$$\varepsilon = -N \frac{d\Phi_B}{dt} = NBA\omega \sin \omega t \quad (10.4.3)$$

If we connect the generator to a circuit which has a resistance R , then the current generated in the circuit is given by

$$I = \frac{|\varepsilon|}{R} = \frac{NBA\omega}{R} \sin \omega t \quad (10.4.4)$$

The current is an alternating current which oscillates in sign and has an amplitude $I_0 = NBA\omega / R$. The power delivered to this circuit is

$$P = I |\varepsilon| = \frac{(NBA\omega)^2}{R} \sin^2 \omega t \quad (10.4.5)$$

On the other hand, the torque exerted on the loop is

$$\tau = \mu B \sin \theta = \mu B \sin \omega t \quad (10.4.6)$$

Thus, the mechanical power supplied to rotate the loop is

$$P_m = \tau \omega = \mu B \omega \sin \omega t \quad (10.4.7)$$

Since the dipole moment for the N -turn current loop is

$$\mu = NIA = \frac{N^2 A^2 B \omega}{R} \sin \omega t \quad (10.4.8)$$

the above expression becomes

$$P_m = \left(\frac{N^2 A^2 B \omega}{R} \sin \omega t \right) B \omega \sin \omega t = \frac{(NAB\omega)^2}{R} \sin^2 \omega t \quad (10.4.9)$$

As expected, the mechanical power put in is equal to the electrical power output.

10.6 Summary

- The **magnetic flux** through a surface S is given by

$$\Phi_B = \iint_S \vec{B} \cdot d\vec{A}$$

- Faraday's law** of induction states that the induced emf ε in a coil is proportional to the negative of the rate of change of magnetic flux:

$$\varepsilon = -N \frac{d\Phi_B}{dt}$$

- The direction of the induced current is determined by **Lenz's law** which states that the induced current produces magnetic fields which tend to oppose the changes in magnetic flux that induces such currents.
- A **motional emf** ε is induced if a conductor moves in a magnetic field. The general expression for ε is

$$\varepsilon = \oint (\vec{v} \times \vec{B}) \cdot d\vec{s}$$

In the case of a conducting bar of length l moving with constant velocity $\vec{v}$ through a magnetic field which points in the direction perpendicular to the bar and $\vec{v}$, the induced emf is $\varepsilon = -Bvl$.

- An induced emf in a stationary conductor is associated with a **non-conservative electric field** $\vec{E}_{nc}$:

$$\varepsilon = \oint \vec{E}_{nc} \cdot d\vec{s} = -\frac{d\Phi_B}{dt}$$

EXAMPLE 1

A conducting rod of length l moves with a constant velocity $\vec{v}$ perpendicular to an infinitely long, straight wire carrying a current I , as shown in the Figure 10.9.4. What is the emf generated between the ends of the rod?

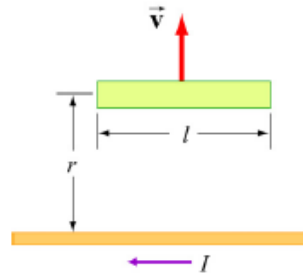


Figure 10.9.4 A bar moving away from a current-carrying wire

SOLUTIONS

From Faraday's law, the motional emf is

$$|\varepsilon| = Blv \quad (10.9.14)$$

where v is the speed of the rod. However, the magnetic field due to the straight current-carrying wire at a distance r away is, using Ampere's law:

$$B = \frac{\mu_0 I}{2\pi r} \quad (10.9.15)$$

Thus, the emf between the ends of the rod is given by

$$|\varepsilon| = \left(\frac{\mu_0 I}{2\pi r} \right) lv \quad (10.9.16)$$

EXAMPLE 2

A conducting rod of length l is free to slide on two parallel conducting bars as in Figure 10.9.3.

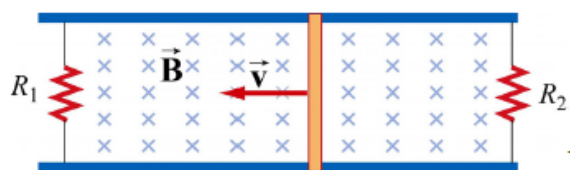


Figure 10.9.3 Sliding rod

In addition, two resistors R_1 and R_2 are connected across the ends of the bars. There is a uniform magnetic field pointing into the page. Suppose an external agent pulls the bar to the left at a constant speed v . Evaluate the following quantities:

- (a) The currents through both resistors;
- (b) The total power delivered to the resistors;
- (c) The applied force needed for the rod to maintain a constant velocity.

SOLUTIONS

(a) The emf induced between the ends of the moving rod is

$$\varepsilon = -\frac{d\Phi_B}{dt} = -Blv \quad (10.9.9)$$

The currents through the resistors are

$$I_1 = \frac{|\varepsilon|}{R_1}, \quad I_2 = \frac{|\varepsilon|}{R_2} \quad (10.9.10)$$

Since the flux into the page for the left loop is decreasing, I_1 flows clockwise to produce a magnetic field pointing into the page. On the other hand, the flux into the page for the right loop is increasing. To compensate the change, according to Lenz's law, I_2 must flow counterclockwise to produce a magnetic field pointing out of the page.

(b) The total power dissipated in the two resistors is

$$P_R = I_1 |\varepsilon| + I_2 |\varepsilon| = (I_1 + I_2) |\varepsilon| = \varepsilon^2 \left(\frac{1}{R_1} + \frac{1}{R_2} \right) = B^2 l^2 v^2 \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \quad (10.9.11)$$

(c) The total current flowing through the rod is $I = I_1 + I_2$. Thus, the magnetic force acting on the rod is

$$F_B = IlB = |\varepsilon| l B \left(\frac{1}{R_1} + \frac{1}{R_2} \right) = B^2 l^2 v \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \quad (10.9.12)$$

and the direction is to the right. Thus, an external agent must apply an equal but opposite force $\vec{F}_{\text{ext}} = -\vec{F}_B$ to the left in order to maintain a constant speed.

10.10 Conceptual Questions

1. A bar magnet falls through a circular loop, as shown in Figure 10.10.1

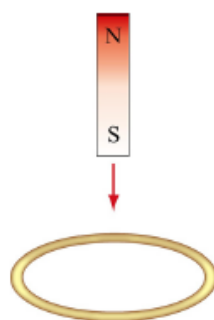


Figure 10.10.1

- (a) Describe qualitatively the change in magnetic flux through the loop when the bar magnet is above and below the loop.
- (b) Make a qualitative sketch of the graph of the induced current in the loop as a function of time, choosing I to be positive when its direction is counterclockwise as viewed from above.
2. Two circular loops A and B have their planes parallel to each other, as shown in Figure 10.10.2.

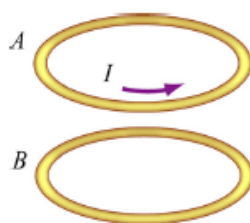


Figure 10.10.2

Loop A has a current moving in the counterclockwise direction, viewed from above.

- (a) If the current in loop A decreases with time, what is the direction of the induced current in loop B ? Will the two loops attract or repel each other?
- (b) If the current in loop A increases with time, what is the direction of the induced current in loop B ? Will the two loops attract or repel each other?
3. A spherical conducting shell is placed in a time-varying magnetic field. Is there an induced current along the equator?